⚔️ INTRODUCTION TO INTEGRALS ⚔️

Calculus • Unit 4 Worksheet • Definite & Indefinite Integrals

Name: _____ Period: _____ Date: _____

📜 KEY FORMULAS – DON'T FORGET THESE!

$\int x^n \, dx = x^{n+1}/(n+1) + C$

$\int k \, dx = kx + C$

$\int e^x \, dx = e^x + C$

$\int \cos(x) \, dx = \sin(x) + C$

$\int \sin(x) \, dx = -\cos(x) + C$

$\int 1/x \, dx = \ln|x| + C$

🔵 SECTION 1: GOJO'S INFINITY – CONCEPTUAL QUESTIONS

Satoru Gojo's "Infinity" technique divides the space between him and threats into infinitely small pieces — just like how integration sums infinitely small slices to find area!

CONCEPT

1. Gojo explains to his students: "An indefinite integral is an anti-derivative." In your own words, explain what it means for one function to be the anti-derivative of another. Give an example.

Write your answer here...

CONCEPT

2. Gojo claims the "+C" in indefinite integrals represents all possible "parallel universes" of a function family. Why must we always include +C?

Your answer...

CONCEPT

3. Evaluate the indefinite integral and explain each step:

$$\int 6x^2 \, dx$$

Show work...

APPLY

4. Gojo divides a battlefield region. The cross-sectional width at position x is f(x) = 3x + 2. Find the general anti-derivative F(x) and verify by differentiating your answer.

Show all work...

🟢 SECTION 2: DR. DOOM'S CALCULATIONS – POWER RULE PRACTICE

Victor Von Doom insists only those who master the Power Rule are worthy of Latveria. Prove yourself.

Find each indefinite integral. Show all steps — Doom demands it.

APPLY

5. $\int x^5 \, dx$

Work & Answer:

APPLY

6. $\int 4x^3 - 7x + 2 \, dx$

Work & Answer:

APPLY

7. $\int \sqrt{x} \, dx$ (Hint: rewrite as x^(1/2))

Work & Answer:

APPLY

8. $\int (1/x^3) \, dx$ (Hint: rewrite as x^{-3})

Work & Answer:

⚡ CHALLENGE

9. Doom's energy cannon outputs power at rate $P(t) = 12t^2 - 6t + 5$ (in megawatts), where t is in seconds. Find the total energy function $E(t) = \int P(t) \, dt$, then find the total energy used from t = 0 to t = 3 seconds.

Show all work...

🔴 SECTION 3: SUPERMAN'S FLIGHT PATH – DEFINITE INTEGRALS

Clark Kent uses definite integrals to calculate exact distances when flying at variable speeds. Remember: $\int_a^b f(x)\,dx = F(b) - F(a)$ (Fundamental Theorem of Calculus)

APPLY

10. Superman's velocity (in km/s) as he approaches Metropolis is $v(t) = 2t + 3$. Calculate the exact distance he travels from $t = 1$ to $t = 4$ seconds.

$\int_1^4 (2t + 3)\,dt$

Show all work using FTC...

APPLY

11. Evaluate the definite integral:

$\int_0^3 x^2\,dx$

Work & Answer:

APPLY

12. Evaluate the definite integral:

$\int_1^4 (3x - 1)\,dx$

Work & Answer:

CONCEPT

13. Superman notes that $\int_2^5 f(x)\,dx = 10$ and $\int_2^8 f(x)\,dx = 22$. Without integrating, what is $\int_5^8 f(x)\,dx$? Explain the property you used.

Answer & Explanation:

🟣 SECTION 4: SUNG JINWOO'S SHADOW ARMY — AREA UNDER THE CURVE

Sung Jinwoo calculates the total mana released by his shadow army over time — the area under a power curve. Arise! 🖤

APPLY

14. Jinwoo's mana output rate is modeled by $M(t) = -t^2 + 6t$ (in units/sec) from $t = 0$ to $t = 6$. Find the total mana released during this interval (the area under the curve).

$\int_0^6 (-t^2 + 6t)\, dt$

Show all work. Include units in your answer.

⚡ CHALLENGE

15. During a dungeon raid, Jinwoo gains experience at rate $E(x) = x^3 - 3x^2 + 4$ (exp/meter). Find the *net* experience gained as he travels from $x = 0$ to $x = 3$ meters. Does he gain or lose net experience? Justify with your calculation.

Show all work...

⚡ **BOSS CHALLENGE**

16. Jinwoo faces the Shadow Monarch. His power $P(x) = 2x$ and the Shadow Monarch's power $Q(x) = x^2$ are graphed on the same axes from $x = 0$ to $x = 2$.

a) Find where $P(x)$ and $Q(x)$ intersect on $[0, 2]$.

b) Set up and evaluate the integral representing the area *between* the two curves on $[0, 2]$. (Hint: $\int$[top − bottom] dx)

c) Who has the advantage in this region, and why?

Show all work for parts a, b, and c...

🌟 SECTION 5: FINAL BOSS ROUND – MIXED REVIEW

APPLY

17. Evaluate:

$\int (5e^x + \cos x)\, dx$

Work & Answer:

APPLY

18. Evaluate:

$\int_0^{\pi} \sin(x)\, dx$

Work & Answer:

CONCEPT

19. Reflection: Match each character to how they connect to integration. Draw lines or write the letter next to the name.

A. Gojo Satoru ___

B. Dr. Doom ___

C. Superman ___

D. Sung Jinwoo ___

1. Summing mana over time = area under curve
2. Infinity as the limit of infinite slices
3. Distance = integral of velocity
4. Power Rule mastery = control over forces

🖤 "I alone level up." — Sung Jinwoo | 🔵 "Throughout Heaven and Earth, I alone am the honored one." — Gojo | 🟢 "Bow before Doom." — Dr. Doom | 🔴 "Truth, Justice, and the integral of velocity." — Superman

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